SAHAYAK Platform: Comprehensive Financial & Cost Architecture

Date: October 2026 | **Base Conversion Rate:** $1 USD = Rs. 96.75

1. Core App Store & Setup Fees (One-Time & Recurring)

- **Google Play Console:** $25 USD one-time setup fee. Converted at Rs. 96.75, this equals approximately **Rs. 2,419**. Note that Google requires a mandatory identity verification track and a strict 14-day closed testing window with at least 12 distinct testers before actual deployment to production.
- **Apple Developer Program:** $99 USD annually, translating to roughly **Rs. 9,578 per year**. Given that the targeted pilot user demographic is located strictly within the North East Region (NER) of India, launching exclusively on Android first is the recommended strategy to bypass this recurring cost during the validation phase.

2. Cloud Infrastructure & Database Maintenance

- **AWS Lightsail Server Tier:** A base Linux instance for handling node logic sits at $5 USD per month (approx. **Rs. 484 per month**) containing 1 vCPU, 1 GB RAM, and 20 GB SSD.
- **AWS Managed Database Tier:** A managed PostgreSQL deployment sits at $15 USD per month (approx. **Rs. 1,451 per month**) for the micro tier. Note that new AWS accounts unlock a 3-month free trial tier on these specific setups, bringing initial trial infrastructure costs down to Rs. 0.
- **Firebase Backend Integration:** The Firebase Spark Plan operates entirely at $0 per month, seamlessly managing standard user authentication paths and automated push notifications. However, if cloud storage pipelines are activated to record voice files, moving to the pay-as-you-go Blaze Plan becomes mandatory.

3. Voice Generation: ElevenLabs Integration Model

Integrating custom human-like voice synthesis for elderly users alters the billing structure fundamentally from per-minute processing to character-based generation volume. A baseline speech framework maps to roughly 1,000 characters generated per minute of interactive audio output.

- **Instant Voice Cloning (IVC):** Requires brief reference audio samples, creating highly realistic local replicas. Available on the Starter Tier starting at $6 USD per month.
- **Professional Voice Cloning (PVC):** Delivers hyper-realistic, deep voice synthesis paths optimized for long-term accessibility, unlocking exclusively on the Creator Tier at $22 USD per month.

4. Financial Projections & Scaling Matrix

The projections below assume an estimated usage framework where each active patient interacts with 2 minutes of generated custom voice assistance per day.

Scale Milestone	Monthly Audio Volume	Character Traffic	Recommended Plan & Overages	Total Estimated Monthly Run Rate
50 Patients (Pilot)	3,000 Minutes	~3 Million Chars	Starter Tier + Overages (~$300)	Rs. 29,025 / month
1,000 Patients	60,000 Minutes	~60 Million Chars	Pro Tier + Overages (~$4,800)	Rs. 4,64,400 / month
10,000 Patients	600,000 Minutes	~600 Million Chars	Enterprise Tier (~$48,000)	Rs. 46,44,000 / month

5. Cost Optimization Framework

- **Audio Caching Optimization:** Rather than making dynamic per-character calls to ElevenLabs API endpoints for repetitive navigation commands, pre-generate fixed UI phrases (e.g., standard greetings, game rules) exactly once. Store these locally inside the application bundle to prevent recurring fees.
- **Backend Unification:** Avoid maintaining parallel backend frameworks. Consolidate systems fully onto either AWS Lightsail (Node.js + PostgreSQL) or Firebase to remove platform cost overlaps.
- **Strategic Regional Partnerships:** Capitalise on existing regional frameworks, noting that Bhashini maintains an established partnership blueprint with the Assam Startup Foundation. This presents a key non-dilutive pathway to leverage localized language infrastructure tools.